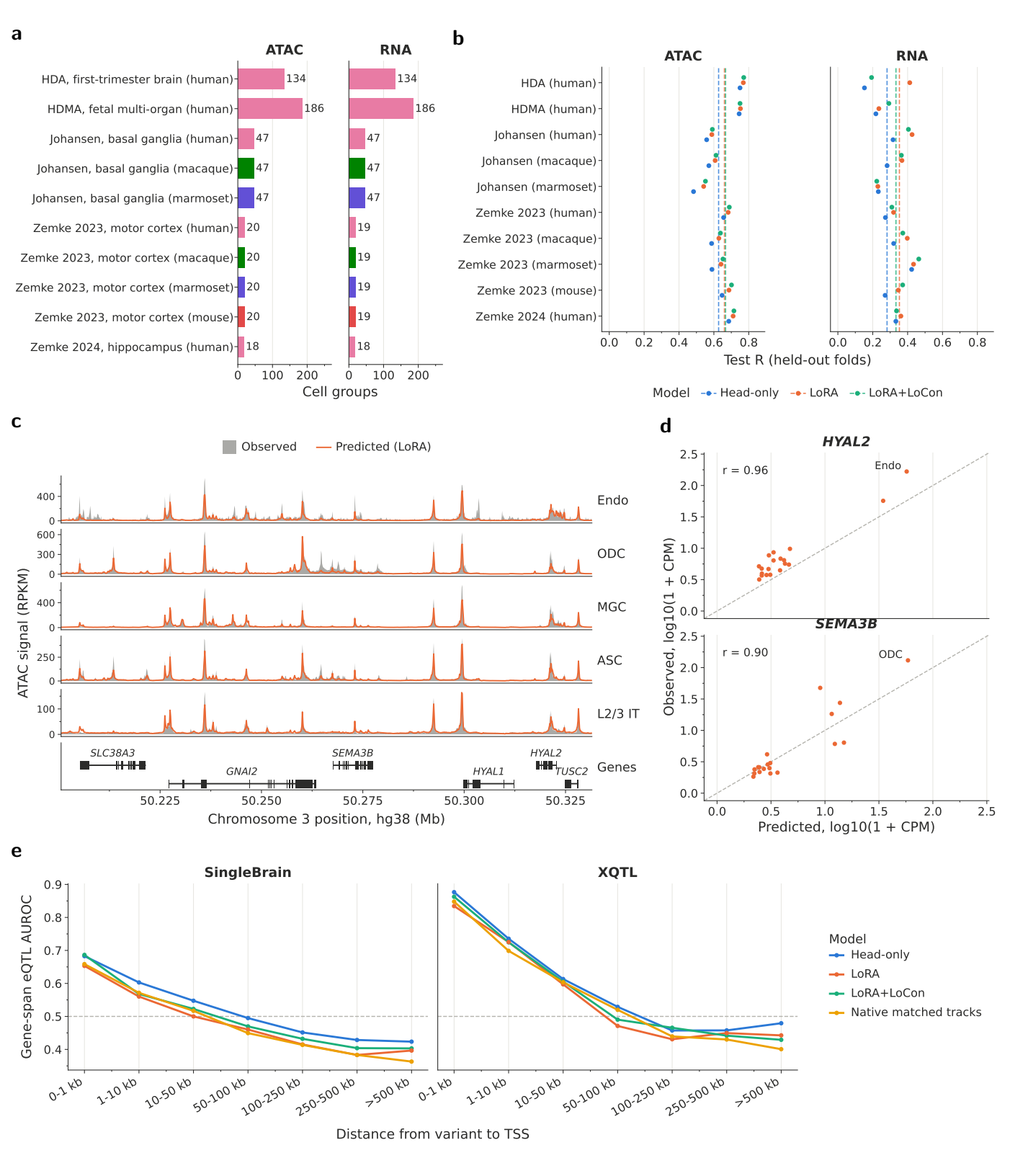

a
ATAC
RNA
HDA, first-trimester brain (human)
HDMA, fetal multi-organ (human)
Johansen, basal ganglia (human)
Johansen, basal ganglia (macaque)
Johansen, basal ganglia (marmoset)
Zemke 2023, motor cortex (human)
Zemke 2023, motor cortex (macaque)
Zemke 2023, motor cortex (marmoset)
Zemke 2023, motor cortex (mouse)
Zemke 2024, hippocampus (human)
134
186
47
47
47
20
20
20
20
134
186
47
47
47
19
19
19
19
18
18
0 100 200
0 100 200
Cell groups
b
ATAC
RNA
HDA (human)
HDMA (human)
Johansen (human)
Johansen (macaque)
Johansen (marmoset)
Zemke 2023 (human)
Zemke 2023 (macaque)
Zemke 2023 (marmoset)
Zemke 2023 (mouse)
Zemke 2024 (human)
0.0 0.2 0.4 0.6 0.8
0.0 0.2 0.4 0.6 0.8
Test R (held-out folds)
Model
Head-only
LoRA
LoRA+LoCon
c
Observed
Predicted (LoRA)
ATAC signal (RPKM)
Endo
ODC
MGC
ASC
L2/3 IT
Genes
SLC38A3
GNAI2
SEMA3B
HYAL1
HYAL2
TUSC2
50.225
50.250
50.275
50.300
50.325
Chromosome 3 position, hg38 (Mb)
d
HYAL2
r = 0.96
Endo
SEMA3B
r = 0.90
ODC
Observed, log10(1 + CPM)
Predicted, log10(1 + CPM)
e
SingleBrain
XQTL
Gene-span eQTL AUROC
0-1 kb
1-10 kb
10-50 kb
50-100 kb
100-250 kb
250-500 kb
>500 kb
Distance from variant to TSS
Model
Head-only
LoRA
LoRA+LoCon
Native matched tracks